

Technical and measurement report

Intra- and inter-rater reliability in ultrasonographic measurements of coracohumeral distance

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ABSTRACT

Background: Ultrasonography (US) is a widely used diagnostic tool in physical therapy. One of the US variables often utilized to quantify the dimensions of the subacromial space and its association with shoulder pathology is the coracohumeral distance (CHD), however, this measurement presents diverse evidence in terms of reliability. **Objectives:** To assess the intra- and inter-rater reliability of both expert and non-expert raters when measuring CHD through US in asymptomatic subjects. Additionally, we determined the effect of rater experience and measurement conditions on the recording of CHD.

Methods: CHD of 15 individuals were recorded from US images of the glenohumeral joint of both upper extremities in three different positions. An expert and a non-expert rater in US usage recorded three CHD measurements, after a randomization procedure. To determine intra- and inter-rater reliability, the interclass correlation coefficient (ICC) and a multivariate variance model for the effects of rater experience, joint position and time of measure were used. Standard Error of Measurement and Minimal Detectable Change was also estimated for CHD measurements.

Results: Intra-rater reliability ranged 0.970 to 0.998) and Inter-rater reliability ranged 0.48 to 0.876). Joint position ($F_{2,55} = 38.308$; $p < 0.001$; $\eta_p^2 = 0.582$) and measurement time ($F_{2,55} = 6.019$; $p = 0.004$; $\eta_p^2 = 0.180$) effect was observed on CHD.

Conclusion: Excellent intra- and poor to moderate inter-rater reliability between expert and non-expert clinicians was determined, the latter being influenced by the position of the glenohumeral joint position at the time of US recording and the time of recording the measurement.

1. Introduction

Ultrasonography (US) is a widely used diagnostic tool in physical therapy. This is primarily due to its convenience, cost-effectiveness, radiation-free nature, and non-invasiveness, making it an ideal method to identify structural alterations in real-time. Specifically, in the shoulder region, US has been employed to quantify changes in the dimensions of the subacromial space, both in sample with shoulder complaints (Oh et al., 2016; Ting et al., 2023) and in asymptomatic sample (Charry et al., 2021; Guerra-Rodríguez et al., 2021). One of the US

variables often used to quantify the dimensions of the subacromial space and its association with shoulder pathology, is the coracohumeral distance (CHD). A decrease of more than 6 mm in the CHD has been described to cause subcoracoid impingement, anterior shoulder pain, and subscapularis tendon tears (Brunkhorst et al., 2013; Misirliloglu et al., 2012; Oh et al., 2016).

While its association with shoulder pathologies has been established, to the best of our knowledge only few studies report reliability values, using interclass correlation coefficient (ICC) and only considering intra-rater agreement for this measure in people with shoulder complaints. In

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this context, [Oh et al. \(2016\)](#) reported ICCs between 0.927 and 0.955 in external, internal and neutral rotation positions of the glenohumeral joint ([Oh et al., 2016](#)); [Navarro-Ledesma et al. \(2017\)](#) showed ICCs of 0.988 and 0.989 for CHD at 0° and 60° elevation ([Navarro-Ledesma et al., 2017](#)). Recently, [Ting et al. \(2023\)](#) reported ICC between 0.893 and 0.996, in maximum external rotation position, in abduction and flexion ([Ting et al., 2023](#)).

Reliability measures had been also shown on asymptomatic subjects. [Charry et al. \(2021\)](#) reported ICC of 0.960 for the 70° elevation position in the scapular plane ([Charry et al., 2021](#)). Subsequently, [Guerra-Rodríguez \(2021\)](#) reported ICC values of 0.740–0.948 for abduction positions with external rotation, zero position and resting positions of the glenohumeral joint ([Guerra-Rodríguez et al., 2021](#)). A commonality among these authors is that CHD measurements were conducted by comparing them against other imaging methods (i.e., magnetic resonance imaging) or by sonographers with expertise in the musculoskeletal area, rather than by individuals with little or no experience.

Since variability in training could potentially impact the psychometric properties of US, the assessment of competence and education becomes an important area of study ([Strike et al., 2023](#)). Notably, there have been no reported studies on the reliability of the CHD measurements between experts and non-experts (i.e., inter-rater reliability), nor on the potential effects that rater experience may have on the magnitude of CHD. Moreover, it has been proposed that research should focus on describing agreement between raters ([Oh et al., 2016](#)). Therefore, the primary aim of this study was to assess the intra- and inter-rater reliability of both expert and non-expert raters when measuring CHD from US images in asymptomatic subjects. Additionally, the secondary aim

was to determine the effect of rater experience and measurement conditions on the measure of CHD.

2. Materials and methods

2.1. Study design

This methodological study was designed in accordance with the Helsinki Consensus on biomedical research in humans, following the Guidelines for Reporting Reliability and Agreement Studies (GRRAS) statement to report reliability and agreement studies ([Kottner et al., 2011](#)). The research protocol was approved by the Bioethics Committee of the Faculty of Rehabilitation Sciences of the Universidad Andrés Bello, and an informed consent form was read and signed by each participant.

2.2. Participants

Participants were recruited following inclusion criteria: age between 18 and 25 years; normal body mass index (BMI); and normal shoulder mobility in both upper extremities. A history of spine, shoulder and elbow trauma, previous shoulder surgery or manipulation under anesthesia, female sex, shoulder pain during the past three months and diagnosed ligament hyperlaxity were considered as exclusion criteria.

2.3. Instruments

A diagnostic ultrasound unit, SonoScape (E2 Digital Color Doppler

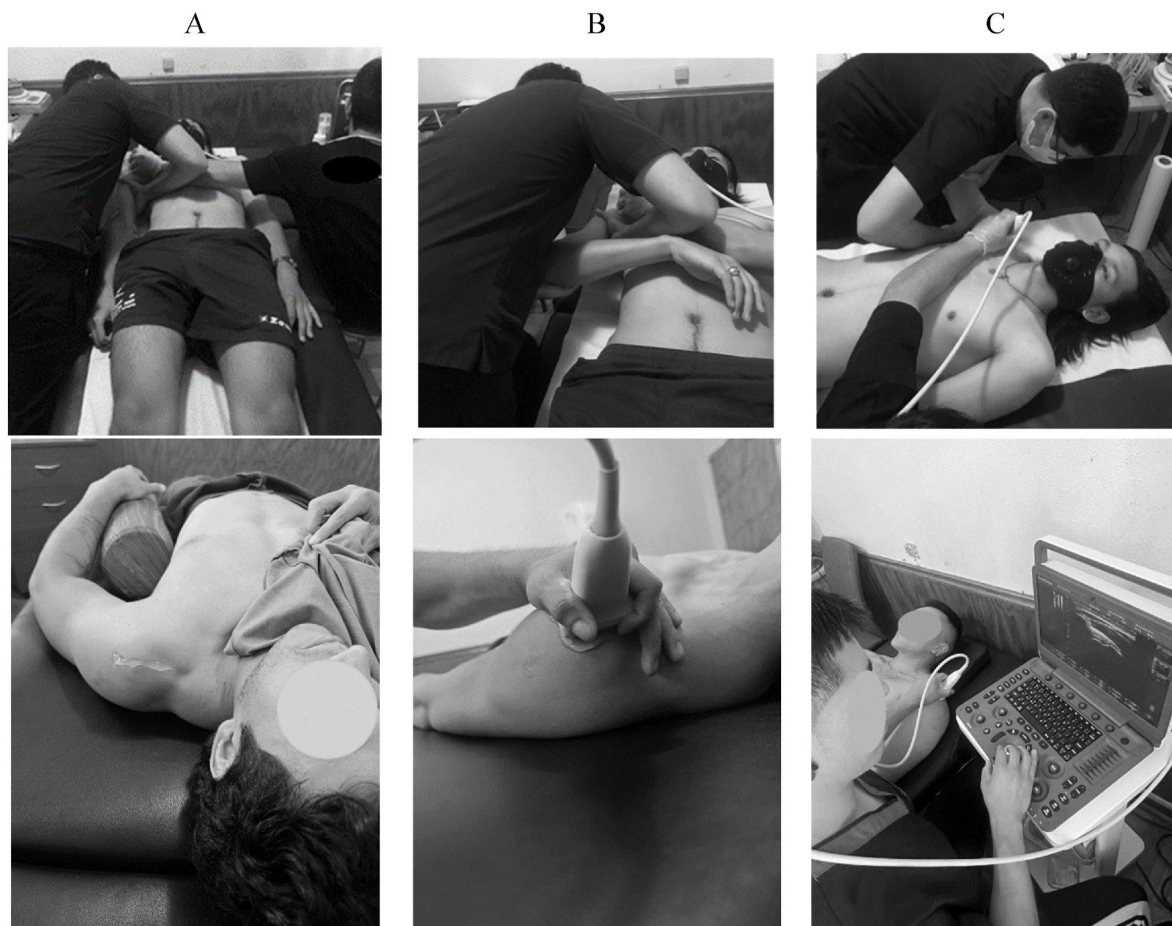


Fig. 1. Execution of distraction techniques (first row) in zero position (column A), resting position (column B) and 90° abduction position with maximum external rotation (column C) and probe position. Extracted and modified with permission from [Guerra-Rodríguez et al. \(2021\)](#). Ultrasonography probe and management is shown on second row (A through B).

Ultrasound System) with a dynamic range of up to 200 dB, was used with a 13 MHz linear transducer with 128 piezoelectric crystals. Images were obtained by a single examiner (radiologist).

2.4. Procedures

Each participant was placed in supine position on a procedure table. Then, participant's glenohumeral joints were positioned in zero position (ZP), resting position (RP), and in 90° of abduction plus maximum external rotation (ABD + ER) position (Fig. 1, first row). ZP corresponds to the location of the arm parallel to the body, with the elbow in extension and thumb in the ventral direction (Kaltenborn, 2004). RP corresponds to an abduction of 25° in the scapular plane (positioning the scapula 30° anterior to the frontal plane) with the elbow flexed and the forearm in the horizontal plane (Lin et al., 2007). For ABD + ER, the glenohumeral joint was positioned at 90° of abduction with maximum external rotation (Oh et al., 2016). The positions were randomized across participants and held by a clinician expert on manual therapy techniques, with 8 years of clinical expertise, a degree in Orthopedic Manual Therapy, head teacher of Biomechanics and Clinical Traumatology classes and research in musculoskeletal area. For all procedures, participants lay supine to ensure scapular stabilization by body weight (Witt and Talbott, 2016). US images were recorded from both upper extremities. The radiologist adjusted US pressure, depth and gain according to participant habitus (Fig. 1, second row). Previous probe position and maintenance of glenohumeral joint position indications were followed (Guerra-Rodríguez et al., 2021). For CHD measure, raters considered the shortest linear distance between the coracoid and the adjacent humeral head (Navarro-Ledesma et al., 2017).

2.5. Raters

Two evaluators were recruited to estimate the CHD from the images obtained by the radiologist. Rater A was a musculoskeletal and sports physical therapist with over 150 h of theoretical and practical training in musculoskeletal US related to diagnostic exploration and intervention and uses this tool in their daily clinical practice. On the other hand, rater B was a physical therapist with manual therapy training, anatomy lecturer with 8 years of experience, and with clinical practice in the musculoskeletal field, with no experience in US imaging.

2.6. Measures

By position, one image was taken (1 image x 3 positions x 15 participants x 2 shoulders = 90 CHD measures). Once images were taken, they were stored in a digital folder. Images were then randomized by a blinded investigator. These randomized images were then loaded back into the US device for evaluation by raters A and B, who independently conducted the first CHD measurements. Subsequently, a technician, blinded to the raters' markings, erased the measurement lines drawn by each rater. Following the erasure of the markings, the images were exported once again from the US device and stored in the digital folder. This sequence of actions was repeated prior to the second and third CHD measurements. Each rater was given one and a half minutes to measure CHD per image during the assessment. Two sessions were utilized for each cycle of measurements.

2.7. Statistical analysis

Sample size calculation was performed in G*Power version 3.1.9.4. (G*Power, Version 3.0.10, Franz Faul, Universität Kiel, Kiel, Germany) (Faul et al., 2007), following previous recommendations (Grissom and Kim, 2014). Data presented by Guerra-Rodríguez et al. (2021) were used for estimating sample size. Considering an effect size of 0.66, a significance level $\alpha = 0.05$ and a power $\beta = 0.95$, a sample of 21 subjects was required. Statistical analysis was carried out in SPSS v21.0 (IBM Corp.,

Armonk, NY, USA). Shapiro-Wilk and Levene tests were used to determine data distribution and homogeneity of variances, respectively. Demographic characteristics were presented as mean and standard deviation (STD).

Reliability assessments were performed using a 2-way random-effects model, absolute agreement, and average measurement (Shrout and Joseph, 1979), considering poor reliability if $ICC < 0.5$; moderate reliability if $0.5 \leq ICC < 0.75$; good reliability if $0.75 \leq ICC < 0.9$; and excellent reliability if $ICC \geq 0.9$ (Koo and Li, 2016). Lower and upper bounds of 95% confidence interval absolute difference (|CI difference|) was also estimated to analyze the rate of change through measurements. ICC procedures were carried out on Matlab R2021b (The MathWorks Inc., Natick, MA, USA) following previous recommendations (McGraw and Wong, 1996).

To determine the effects of measurement conditions on CHD, a multivariate analysis of variance was applied. CHD measurement time (3: measurement 1; measurement 2; measurement 3) and distraction positions (3: ZP; RP; ABD + RE) were considered as intrasubject factors and rater (2: rater A; rater B), and hemibody (2: right hemibody; left hemibody) as intersubject factors. Multivariate and univariate effects were analyzed. In case of significant effects, the partial eta-squared statistic (η_p^2) was used for the purpose of examining the size of the effect, being interpreted as a small effect if $0.06 > \eta_p^2$; moderate effect if $0.06 \leq \eta_p^2 < 0.14$; and great effect if $\eta_p^2 \geq 0.14$ (Cohen, 1992). Finally, in case of significant effects, multiple comparisons with Tukey's *post hoc* correction were applied. Standard error of measurement ($SEM = \sqrt{1 - reliability} \times STD$) and minimal detectable change ($MDC = 1.96 \times \sqrt{2} \times SEM$) were also calculated. The measures carried out by each rater were also described. For all statistical tests, an $\alpha = 0.05$ was used.

3. Results

3.1. Participants

Fifteen male participants were recruited (age = 20.93 ± 0.46 years; weight = 71.06 ± 3.16 kg; height = 1.73 ± 0.02 m; BMI = 23.84 ± 1.04 kg/m²). Table 1 outlines the CHD measurements conducted by raters in different positions, presenting the corresponding SEM and MDC values.

Table 1
Coracohumeral distance measures.

Rater	Position (Measure)	Mean (Std)	SEM	MDC
A	ZP (1)	16.42 (4.06)	1.05	3.18
	ZP (2)	16.59 (3.98)	1.02	3.12
	ZP (3)	16.60 (4.09)	1.05	3.20
	RP (1)	13.09 (4.77)	1.23	3.73
	RP (2)	13.16 (4.77)	1.23	3.73
	RP (3)	13.19 (4.81)	1.24	3.76
	ABD + ER (1)	15.96 (2.94)	0.76	2.30
	ABD + ER (2)	16.22 (2.84)	0.73	2.23
	ABD + ER (3)	16.35 (2.91)	0.75	2.27
B	ZP (1)	17.39 (4.68)	1.21	3.67
	ZP (2)	17.40 (4.77)	1.23	3.73
	ZP (3)	17.54 (4.66)	1.20	3.65
	RP (1)	13.27 (4.48)	1.15	3.50
	RP (2)	13.54 (4.87)	1.25	3.81
	RP (3)	13.71 (4.83)	1.24	3.78
	ABD + ER (1)	17.08 (2.77)	0.71	2.17
	ABD + ER (2)	17.25 (2.95)	0.76	2.31
	ABD + ER (3)	17.30 (2.93)	0.75	2.30

Abbreviations: A, US expert rater; B US non-expert rater; ZP, zero position; RP, resting position; ABD + ER, 90° abduction with maximum external rotation position; Std, Standard deviation; SEM, Standard error of measurement; MDC, Minimal detectable change.

3.2. Intra- and inter-rater reliability

Table 2 shows intra-rater reliability ICCs. Both raters reached excellent intra-rater reliability. Table 3 shows the ICCs for inter-rater reliability. Poor and moderate inter-rater reliability was observed for the raters' measurements of CHD in ZP and good reliability for RP and ABD + ER. Regarding the absolute difference in the ICC values ($|\text{CI difference}|$), ZP and RP showed lower magnitudes ($|\text{CI difference}| = 0.843$ to $|\text{CI difference}| = 0.769$ and $|\text{CI difference}| = 0.369$ to $|\text{CI difference}| = 0.200$, respectively), not so in the ABD + ER position, in which an increase is appreciated towards the last measurement ($|\text{CI difference}| = 0.454$ to $|\text{CI difference}| = 0.500$).

3.3. Multivariate analysis of variance

Multiple comparisons are shown in Fig. 2. An effect of the distraction position ($F_{2,55} = 38.308$; $p < 0.001$; $\eta_p^2 = 0.582$) and number of measurement ($F_{2,55} = 6.019$; $p = 0.004$; $\eta_p^2 = 0.180$) was observed on CHD. The rest of the effects were not significant.

4. Discussion

The main objective of this investigation was to determine the intra- and inter-rater reliability of expert and non-expert raters while measuring CHD in asymptomatic subjects using US. Our results suggest that the reliability of both raters is excellent (Table 2), however, between them ranges from poor to moderate (Table 3). Regarding intra-rater reliability, our results are in line with previous evidence, reaching ICCs interpreted as excellent reliability. Moreover, to the best of our knowledge, no research on SEM and MDC values for CHD measurements under different joint positions and distraction techniques has been conducted. In this context, only Navarro-Ledesma et al. (2017) presented SEM and MDC values but for CHD measurements at static positions of 0 and 60° (Navarro-Ledesma et al., 2017). Finally, CHD measurement magnitudes (Table 1 and Fig. 1) also align with previous findings.

To delve deeper into the factors that could influence the differences in magnitudes, we proposed, as a secondary objective, to determine the effects of the rater's experience on the magnitude of CHD in different glenohumeral joint positions. Given that the ICCs are derived from a repeated analysis (i.e., three measurements), the multivariate effects analysis on CHD showed a significant and large effect of position and measurement on CHD, but not of the rater or the participant's hemi-body. While considering CHD magnitudes, significant differences between the first and third measurements can be evidenced in all positions. Although the US images were recorded between techniques, the increase in CHD magnitude may be due to an increase in the laxity of the capsule and ligaments of the glenohumeral joint, as a result of the joint movement due to the applied distraction techniques. This was already discussed by Guerra-Rodríguez et al. (2021) and earlier by Bialosky et al. (2009). Regarding the effects of position, this may be due to the bony eminences changing their orientation when repositioned or due to anatomical variants, an aspect that was discussed by Navarro-Ledesma et al. (2017).

Non-significant effect of the rater on the measurements suggests that

Table 2
Intra-rater interclass correlation coefficients.

Rater	ZP	RP	ABD + ER
Rater A	0.997 [0.995; 0.998]	0.998 [0.997; 0.999]	0.987 [0.976; 0.993]
Rater B	0.997 [0.995; 0.998]	0.970 [0.945; 0.984]	0.996 [0.996; 0.993]

Abbreviations: A, US expert rater; B US non-expert rater; ZP, zero position; RP, resting position; ABD + ER, 90° abduction with maximum external rotation position. Data presented as ICC [lower bound 95% confidence interval; upper bound 95% confidence interval].

Table 3
Inter-rater interclass correlation coefficients.

Position	Measure	ICC [LB; UB]	$ \text{CI difference} $
ZP	1st.	0.481 [-0.090; 0.753]	0.843
	2nd.	0.507 [-0.034; 0.765]	0.800
	3rd.	0.526 [0.004; 0.774]	0.769
RP	1st.	0.772 [0.522; 0.891]	0.369
	2nd.	0.864 [0.714; 0.935]	0.220
	3rd.	0.876 [0.741; 0.941]	0.200
ABD + ER	1st.	0.720 [0.412; 0.866]	0.454
	2nd.	0.751 [0.478; 0.881]	0.403
	3rd.	0.691 [0.353; 0.853]	0.500

Abbreviations: ZP, zero position; RP, resting position; ABD + ER, 90° abduction with maximum external rotation position. Data presented as ICC [lower bound 95% confidence interval; upper bound 95% confidence interval]. $|\text{CI difference}|$ stands as absolute difference between lower bound and upper bound of the 95% confidence interval for interclass correlation coefficients (ICC).

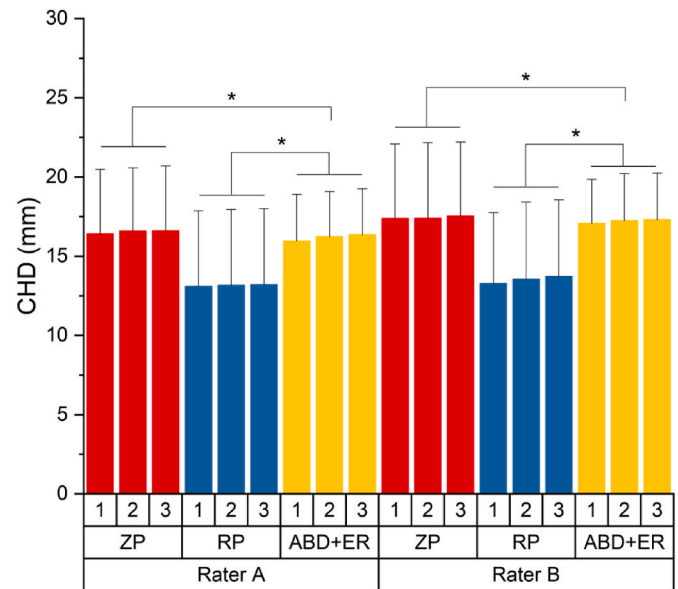


Fig. 2. Multiple comparisons of means by rater, position (ZP, zero position; RP, resting position; and ABD + ER, 90° abduction and maximal external rotation position) and measurement (1, 2, and 3) of coracohumeral distances (CHD) in millimeters. * Indicates statistically significant differences between first and last measure by position and between positions.

the exploration of CHD with US is similar when comparing expert and non-expert raters. It should also be noted that, although the reliability values of the measures by position did not reach an excellent reliability, the difference between the confidence intervals (Table 3) shows a tendency to decrease in the amplitude of this measure of dispersion, which suggests that there is a learning process by the raters with the passing of the CHD estimation Strike et al. (2023).

This research has several limitations. Only young, healthy subjects with inclusion and exclusion criteria were considered, using a non-probabilistic method by convenience, so the results cannot be extrapolated to individuals with shoulder mobility restrictions. The major limitation of US is the dependence on operator skill, by not knowing the normal/abnormal anatomy, or errors caused by the US technique, however, the images were captured by a sonographer expert in exploration of musculoskeletal structures. In addition, only men were considered considering the size of the glenoid cavity and estrogen production during the menstrual cycle, recommending the selection of women for future research. Another limitation is that the analysis of intra- and inter-rater reliability (i.e., expert, and non-expert) was performed using the same image extracted by an independent expert in

ultrasound. Future research should consider intra- and inter-rater reliability by adding the image acquisition process, providing greater clinical value to ultrasound exploration in the shoulder region. Finally, a protocol for measuring CHD is suggested to improve inter-rater reliability.

5. Conclusion

Intra- and inter-rater reliability was determined in the recording of coracohumeral distance magnitudes in asymptomatic subjects in different joint positions. We can conclude that there is an excellent intra-rater reliability and poor to moderate inter-rater reliability, determined by the repetition in the measurements and by the articular position.

CRedit authorship contribution statement

Diego Guerra-Rodríguez: Conceptualization, Funding acquisition, Project administration, Supervision, Validation, Writing – original draft. **Juan Guerrero-Henriquez:** Conceptualization, Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Daniel Basilio:** Methodology, Resources, Software, Visualization, Writing – review & editing. **Guillermo Mendez-Rebolledo:** Investigation, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

Juan Guerrero-Henríquez is CEO of GoStats! E.I.R.L.

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